

Stress Detection Through Sleep-Based Physiological Signals: An Integrated Semi-Supervised and Supervised Learning Approach

Jittapatana Prayoonpruk*, Maxwell Kozlov*, Brandon Ismalej*, Abhinav Neelam*, Xunfei Jiang*, Nhut Ho†

*Department of Computer Science, California State University, Northridge, CA 91330

†Department of Mechanical Engineering, California State University, Northridge, CA 91330

{jittapatana.prayoonpruk.119, maxwell.kozlov.595, brandon.ismalej.671, abhinav.neelam.591}@my.csun.edu, xunfei.jiang@csun.edu, nhut.ho.51@csun.edu

Abstract—Stress significantly affects physical and mental health, manifesting itself in indicators such as elevated heart rate, reduced variability of heart rate, and poor sleep quality. Wearable devices monitor these markers but face challenges with multimodal sensor input processing and generalization due to limited labeled data and individual variability. This study introduces an integrated stress detection framework using semi-supervised learning for data refinement and supervised learning for stress detection. A public sleep heart health dataset was processed through feature selection and transformations, and a self-training classifier enhanced label quality, mitigating the impact of limited labeled data. Our machine learning models, including Random Forest, Decision Tree, and XGBoost, achieved up to 99% accuracy and a 97% F1-score. These results demonstrate the reliability of combining semisupervised and supervised learning for improved stress detection.

Index Terms—stress detection, stress through sleep, supervised learning, semi-supervised learning

I. INTRODUCTION

Stress has become a major public health concern due to its significant impact on both physical and mental health. Prolonged stress has been associated with higher heart rates at night, which, when combined with exposure to early life trauma, can lead to increased anxiety and poor mental health [1]. Chronic stress has also been associated with decreased heart rate variability (HRV), an important physiological parameter associated with cardiovascular risk and poor health outcomes [2]. Studies also have shown that mental stress affects HRV while increasing heart rate, emphasizing the physiological impact of stress on cardiovascular health [3]. Furthermore, poor sleep quality and reduced HRV have been associated to depressive symptoms in people exposed to chronic stress, highlighting the interdependence of stress, sleep, and mental health [4].

Wearable devices, such as smartwatches and fitness trackers, have emerged as promising tools for ongoing health monitoring. These sensors can monitor physiological signals such as heart rate and heart rate variability, which have been shown to be reliable markers of mental stress [5]. In addition, the National Sleep Foundation has emphasized the vital role

of regularity in sleep in maintaining mental health, particularly the association between stress and interrupted sleep patterns [6]. Despite advances in wearable technology, current stress detection systems face significant challenges. These include noise and inconsistencies in the physiological data collected from wearable sensors, limited availability of labeled data for model training, and the complexity of accurately linking physiological signals to stress levels, which can lead to inconsistent model performance and limited generalizability.

This study proposes an integrated stress detection framework combining semi-supervised learning and supervised learning algorithms for detecting stress using physiological data. The pre-processing phase consists of stress survey label refinement using a semi-supervised learning technique with a self-training classifier, as well as feature selection and mathematical transformations to prepare the data for model training. Supervised learning models are then used to accurately detect stress patterns. The supervised learning models are then applied to detect stress patterns effectively. The results of this study demonstrate high accuracy and reliability in stress prediction, with models such as Random Forest, Decision Tree, and XGBoost achieving up to 97% F1-score and 99% accuracy.

This project improves the reliability of stress detection and highlights the value of combining several data processing approaches for better stress monitoring results. The main contributions of this study are:

- Developing a comprehensive supervised learning framework for stress detection using smart wearable data;
- Integrating semi-supervised learning for stress label refinement during data pre-processing, and solving the issue of limited stress labels in the dataset;
- Evaluating the models using key performance metrics such as accuracy, precision, recall, and F1-score.

The rest of this paper is organized as following: Section II discusses related work; Section III presents the framework of the project; Section IV discusses the methodologies that will be used for this project; Section V presents the experimental results; and Section VI concludes the project and discusses

future work.

II. RELATED WORKS

This section provides an overview of research on stress detection utilizing wearable sensors and machine learning approaches. The examined literature is grouped according to the focus of the approaches utilized in stress detection, including stress detection through sleep data analysis and stress detection using wearable devices and sensors.

A. Stress Detection through Sleep Data Analysis

Rahman et al. proposed a method for predicting stress levels early based on sleep data, with the goal of detecting stress as a serious health risk using physiological signs such as heart rate and sleep patterns [7]. Their solution used supervised learning algorithms on the publicly available SaYoPillow dataset [8] [9] which was collected from a Smart Pillow that was connected to a wireless tracker to monitor sleep habits. They concluded that Gaussian Naive Bayes (GNB) and Support Vector Machine (SVM) were effective in predicting stress levels with high accuracy and reliability, obtaining an accuracy score of 100%. However, the dataset has only 630 records, which was small and may have influenced the generalizability of the results.

Jayawickrama and Rupasingha explored stress detection using sleeping habits, emphasizing its relationship between sleep quality and stress levels [10]. They reported using a publicly available dataset from Kaggle although the specific dataset used was not disclosed in the paper. This paper provides an important reference for stress detection using machine learning and physiological data. They trained a variety of machine learning models, including Naive Bayes, Support Vector Machine (SVM), Decision Tree, and Random Forest. Their experimental results showed that Naive Bayes accurately classified stress levels based on sleep data, with a claimed accuracy of 91%, indicating its efficacy in stress classification.

Vardhan et al. studied the use of artificial intelligence for stress detection via sleep analysis, emphasizing the importance of non-invasive and accurate stress monitoring systems based on physiological data [11]. This study also uses the SaYoPillow dataset. Their method includes gathering and pre-processing sleep data for feature extraction before employing a variety of machine learning techniques, including Decision Tree, Support Vector Machine (SVM), AdaBoost, and Multi-Layer Perceptron (MLP). The research claimed that AI-based models, particularly the Multi-Layer Perceptron, were useful for forecasting stress levels with the MLP attaining a 100% accuracy score, closely followed by the Support Vector Machine (SVM) at 99%.

B. Stress Detection Using Wearable Devices and Sensors

Gedam and Paul conducted a comprehensive review of mental stress detection using wearable sensors and machine learning approaches, motivated by the growing demand for non-invasive, continuous stress monitoring to address stress-related health conditions [12]. They classified common physiological indicators for stress detection, including heart rate,

electrodermal activity (EDA), and respiration rate. Their review compared the performance of various machine learning algorithms, such as Support Vector Machines (SVM), Decision Trees, and Neural Networks, in stress classification. They stated that combining several physiological signals with machine learning approaches improves stress detection accuracy. This review provides a well-rounded analysis by incorporating a variety of sensor types and algorithms.

Vos et al. investigated generalizable machine learning models for stress monitoring using wearable sensors, highlighting the importance of accurate stress detection via physiological data [13]. They performed a thorough literature review, examining physiological signals across multiple datasets, including WESAD, SWELL, and AffectiveROAD, chosen for their multimodal physiological signals that are crucial for stress detection. While wearable sensors show potential for stress detection, they pointed out that variation in datasets and feature engineering approaches can have an impact on model performance. They emphasized the need for uniform data collection and model evaluation techniques to improve consistency. This review gives useful insights on generalizable stress detection with wearable technologies.

Lazarou and Exarchos analyzed stress prediction using real-time physiological data analysis in wearable devices, underlining the growing need for reliable stress monitoring due to increased physical and mental health problems [14]. They examined real-time stress detection models based on wearable sensors, with an emphasis on physiological indications such as heart rate variability (HRV), electrodermal activity (EDA), and breathing patterns. The study categorized commonly used wearable sensors and assessed their efficacy for stress detection. Several machine learning models, including Random Forest, Support Vector Machines (SVM), and Convolutional Neural Networks (CNN), were compared in terms of stress categorization accuracy. They found that wearable technology paired with machine learning has great potential to detect stress. However, data inconsistency and personal variability were identified as important problems that need further research.

III. DESIGN

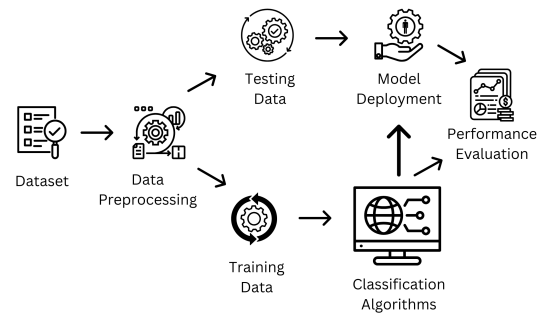


Fig. 1. Stress Detection Architecture

As shown in Fig. 1, this project is composed of five components: dataset, data preprocessing, classification algorithms, performance evaluation, and model deployment. During data preprocessing, raw data is cleaned, converted, and prepared for model training. The classification algorithms are trained on preprocessed data using a variety of machine learning models. The performance evaluation uses testing data to determine the effectiveness of the trained models.

A. Data Pre-processing

The data pre-processing phase is critical in preparing the dataset for effective machine learning models. Fig. 2 shows the pipeline for processing the raw data.

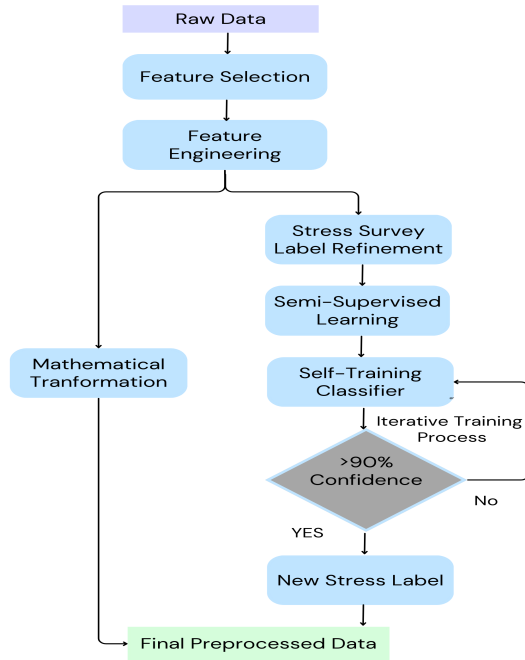


Fig. 2. Data Pre-processing Design

1) *Feature Selection*: This phase aimed to reduce the dataset's dimensionality and align with the objective of stress detection using wearable devices. The feature selection technique used a structured approach to determine the most significant variables while keeping data simple and practical. A large number of features from the dataset were evaluated. To streamline this process, statistical methods such as correlation analysis and mutual information were used to identify the importance of each attribute for stress detection. Furthermore, domain knowledge was used to prioritize parameters recorded by wearable devices, such as heart rate, arousals, sleep hours, sleep stages, and sleep efficiency. Features that require specialized equipment or invasive measurements are omitted to ensure the model's applicability to real-world scenarios. This selection technique maintained a balance between model performance and data availability, focusing on characteristics that could be routinely collected in everyday use.

2) *Feature Engineering*: This stage involves two parallel processes to enhance the dataset.

Mathematical transformation: a Mathematical transformation is used to normalize the data and reduce the effect of outliers, resulting in a more balanced distribution. The mathematical transformation is stated in Eq. (1)

$$x' = -\log(-\log(x + 0.001)) \quad (1)$$

where:

- x is the original data value,
- x' is the transformed data value,
- The constant 0.001 is added to avoid undefined values when $x = 0$.

Stress Survey Label Refinement: this step includes using semi-supervised learning [15] to refine the dataset with limited stress labels and improve its predictive capacity. Because the existing stress survey data lacked accuracy in predicting stress effects, it is insufficient for training a fully supervised model. In order to get around this restriction, a self-training classifier is introduced in a semi-supervised learning algorithm. This classifier iteratively processes the data, assigning stress labels and determining confidence levels. The iterative training continues until the model reaches a confidence level greater than 90%. Once the classifier achieves a confidence level above 90%, new stress labels are generated and integrated into the dataset. The refined stress labels, along with the mathematically transformed data, are then combined to produce the final preprocessed dataset.

B. Machine Learning Model Training and Testing

In this step, the pre-processed dataset is used to train several classification models and evaluate their performance with the testing data. The goal is to find effective models for stress detection using important assessment measures such as accuracy, precision, recall, and F1 score.

1) *Dataset Splitting*: The dataset was divided into two subsets: training data and testing data. The training data was used to create and optimize the machine learning models, whilst the testing data was utilized to assess the model's generalization performance independently. The data was split with 70% for training and 30% for testing.

2) *Supervised Learning Algorithms*: Several classification algorithms were explored for the stress detection task, consisting of K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, Random Forest, Decision Tree, and XGBoost. The purpose of comparing multiple models is to illustrate the superiority of our proposed models. Section IV discusses in detail about the rationale for selecting these algorithms. Each algorithm was tuned using hyperparameter optimization techniques to achieve optimal performance.

3) *Model Training and Testing*: During the training phase, each algorithm was applied to the training dataset. After training, the models were tested on the testing dataset. Performance metrics including accuracy, precision, recall, and F1-score will be calculated for each model.

IV. METHODOLOGIES

To classify the stress level from sleep data, we employed conventional machine learning algorithms, including KNN, Naive Bayes, SVM, Random Forest Classifier, Decision Tree, and XGBoost. In this section, we introduce the key algorithms used for in stress detection.

A. K-Nearest Neighbors

K-Nearest Neighbors (KNN) was selected because it is a simple, non-parametric classifier that compares the distance between new and existing data points. It allocates the new point to the most common class among its 'k' nearest neighbors. KNN performs best with well-separated data, but it is sensitive to the 'k' and feature scaling parameters.

A study employing KNN for stress detection in a free-living setting obtained 98% accuracy, demonstrating its reliability [16]. The KNN algorithm also applies to stress detection during sleep using similar physiological data. However, KNN may struggle with datasets where stress-related physiological features overlap significantly.

B. Naive Bayes Classifier

Naive Bayes is a probabilistic machine learning technique based on Bayes' Theorem that computes the likelihood of a class based on the distribution of feature values. It assumes that the categorization features are independent of one another. Naive Bayes is used for stress detection because to its simplicity, speed, and efficacy with high-dimensional data. It performs well with small datasets and can handle both category and numerical features, making it appropriate for stress prediction using physiological indicators [17]. Its probabilistic nature also yields visible data, which might be valuable in health monitoring and decision-making applications. However, the assumption of feature independence may not always be valid, particularly when dealing with related health indicators. The Naive Bayes formula is expressed as shown in Eq. (2)

$$P(C|\mathbf{x}) = \frac{P(C)P(\mathbf{x}|C)}{P(\mathbf{x})} \quad (2)$$

where:

- $P(C|\mathbf{x})$ is the posterior probability of class C given the features $\mathbf{x}$.
- $P(C)$ is the prior probability of class C .
- $P(\mathbf{x}|C)$ is the likelihood of the features given the class C .
- $P(\mathbf{x})$ is the evidence or total probability of the features $\mathbf{x}$.

C. Support Vector Machine

The Support Vector Machine (SVM) works by identifying the appropriate hyperplane for separating data points into classes. It recognizes support vectors, which are data points that are closest to the decision boundary, and uses them to optimize the margin between classes, thus enhancing classification accuracy. Fig. 3 provides a conceptual illustration of the

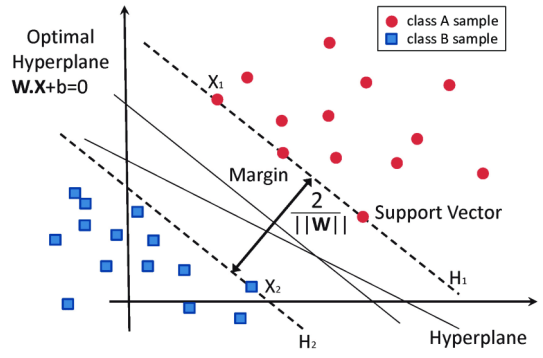


Fig. 3. A Conceptual Illustration of the SVM Algorithm [18]

SVM algorithm, showing the separation of data points using the optimal hyperplane and support vectors.

SVM is a reliable stress detection algorithm, especially when combined with physiological data. However, its limits with huge datasets and the necessity for thorough hyperparameter adjustment indicate that it is best suited for moderate-sized data and structured contexts [12].

D. Decision Tree

Decision Tree was employed because of its ease of use and performance with tabular data. It operates by recursively partitioning the information depending on feature thresholds, making it effective for detecting stress patterns using physiological markers such as heart rate and sleep efficiency [12]. Fig. 4 is an example of a decision tree structure, with the

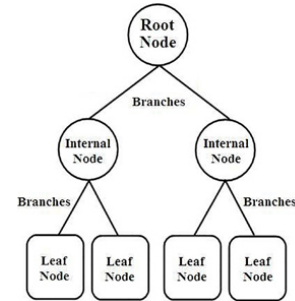


Fig. 4. An Example of Decision Tree [19]

root node, internal nodes, and leaf nodes representing the hierarchical decision-making process.

E. Random Forest Classifier

The Random Forest Classifier was used to assess the effectiveness of ensemble learning. Combining numerous decision trees and averaging their predictions decreases overfitting while increasing the model's generalizability. This approach effectively detects stress by assessing a variety of sleep variables, including heart rate variability and sleep stages, over various decision routes [20].

F. XGBoost

XGBoost, a gradient boosting algorithm, was used to improve model performance by iteratively correcting faults from prior rounds. It employs an ensemble of weak decision trees, with each new tree attempting to minimize the errors generated by the prior trees. This technique is useful for stress detection because it can handle complex data patterns and feature interactions in sleep data [21].

$$y_i = \sum_{k=1}^n f_k(x_i), \quad f_k \in F \quad (3)$$

where:

- F is the set of regression trees,
- f_k denotes a tree,
- y_i denotes the projected value for the i th occurrence of x_i .

In contrast to accepting the final output of Random Forest that won the majority of votes, as shown in Eq. (3), the anticipated output of XGBoost is the sum of all the outputs.

V. RESULTS

The model training process is illustrated in Fig. 5, providing a visual representation of the evaluation workflow.

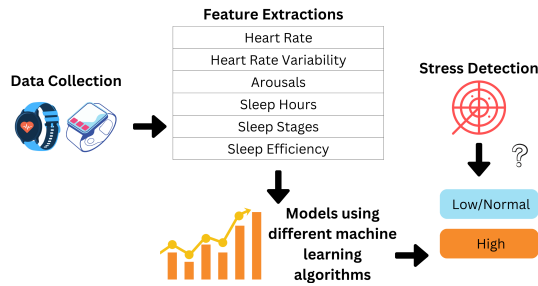


Fig. 5. A Process of Getting Results

A. Dataset

This study utilized the publicly available Sleep Heart Health Study (SHHS) Visit 1 dataset [22] [23] that was collected from 6,441 men and women aged 40 years and older. The dataset was obtained during an extensive polysomnography (PSG) assessment conducted on a single night. It comprises a number of physiological features that can be used to detect stress. The dataset was chosen for its thorough coverage of stress-related physiological signals making it appropriate for testing machine learning models in stress classification tasks.

Table I summarizes the characterization of stress levels using major physiological measures such as heart rate, heart rate variability (SDNN), sleep efficiency, and arousals. These levels are consistent with previous research that found a link between increased heart rate, decreased heart rate variability, low sleep efficiency, and stress-induced physiological changes. The table efficiently distinguishes between healthy (low stress) and harmful (high stress) states, supporting previous research findings on the physiological effects of stress on sleep-related measures.

TABLE I
STRESS LEVEL CHARACTERIZATION TABLE

Sleep (hrs)	Heart Rate (bpm)	Heart Rate Variability (ms)	Sleep Efficiency (percentage)	Arousals (count)	Stress State
7-9	46-89	30-144	47-98	3-55	Low/Normal (Healthy)
2-6	68-99	24-102	59-79	7-52	High (Unhealthy)

B. Feature Selection

Feature selection was critical in optimizing the dataset so that the models captured the most relevant physiological signals for stress detection. Nine significant factors were selected for model training, including instantaneous heart rate (IHR), sleep efficiency, Heart Rate Variability (SDNN), total sleep duration, arousals, percentage of sleep in stages 1, 2, 3 or 4, and REM [24]. These features were chosen because they are relevant to wearable devices and have an established association with stress levels.

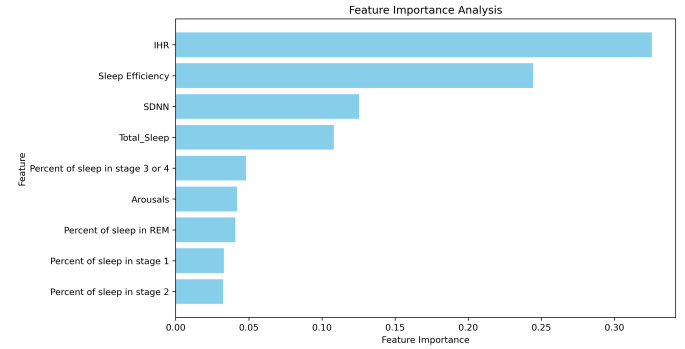


Fig. 6. Feature Importance Analysis

The Random Forest algorithm was used to prioritize feature importance, as seen in Fig. 6. The analysis identifies IHR and Sleep Efficiency as the most important determinants of stress. These characteristics have a significant impact on model performance, with SDNN and Total Sleep being moderately important. The other characteristics, such as arousals and sleep stage distributions, were shown to be less impactful but remain beneficial for improving the model's robustness.

A correlation heatmap, as demonstrated in Fig. 7, was created to investigate the correlations between characteristics and stress labels. The results show a moderate positive connection between IHR and the stress label, consistent with previous research linking higher heart rate to stress. Similarly, Sleep Efficiency showed a significant negative connection, highlighting its usefulness as a stress signal. Other measures, like SDNN and Total Sleep, showed weaker but still significant relationships with stress.

To expand on the link between essential aspects, a joint plot of Sleep Efficiency and IHR against the stress label was constructed in Fig. 8. The scatterplot reveals distinct clusters that separate stressed and non-stressed individuals. Higher

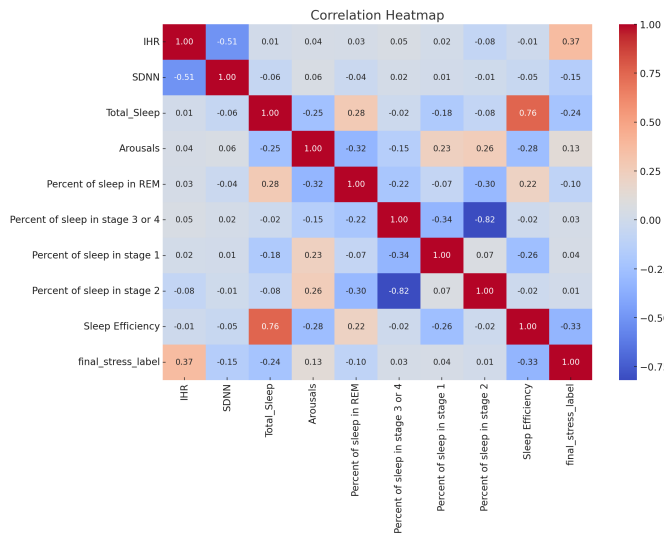


Fig. 7. Correlation Heatmap

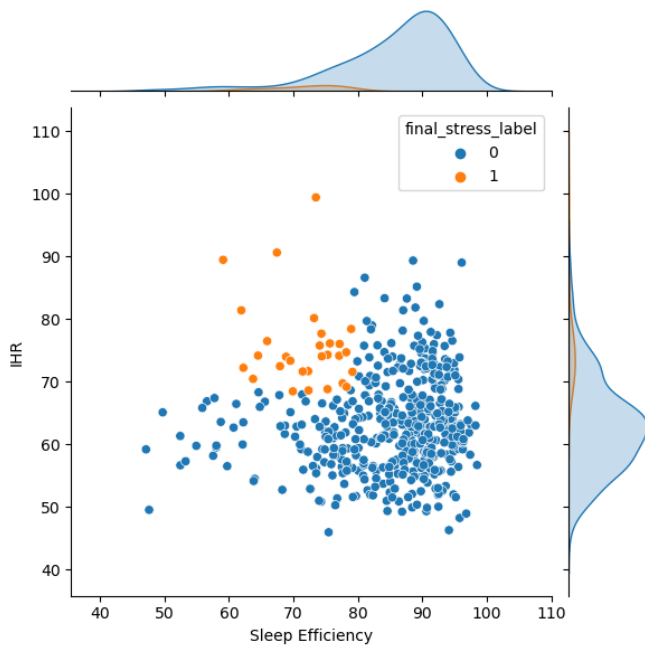


Fig. 8. Join plot of Sleep Efficiency vs. IHR against the stress label

stress levels are connected with lower sleep efficiency and higher IHR, indicating that they have a combined predictive power in stress detection.

Following data analysis and preparation, the final dataset contains approximately 500 records.

C. Model Training

Six machine learning models used for stress detection are analyzed in this section. Performance metrics such as Accuracy, F1 Score, Recall, and Precision were analyzed to determine the effectiveness of these models.

TABLE II
CLASSIFICATION PERFORMANCE RESULTS

Model	Accuracy	F1	Recall	Precision
KNN	0.92	0.48	0.50	0.46
SVM	0.94	0.64	0.60	0.73
Naive Bayes	0.96	0.80	0.77	0.84
Random Forest	0.99	0.97	0.94	1.00
Decision Tree	0.99	0.97	1.00	0.95
XGBoost	0.99	0.97	1.00	0.95

The comparative results, as demonstrated in Table II, highlight significant variations in the performance of the machine learning models tested. XGBoost, Random Forest, and Decision Tree performed better on all four metrics. Each of these models attained an accuracy of 0.99 along with consistently high F1 scores, recall, and precision values, demonstrating their dependability and robustness in stress classification tasks. K-Nearest Neighbors (KNN) yielded the lowest performance among the models, with an accuracy of 0.92, F1 score of 0.48, recall of 0.50, and precision of 0.46.

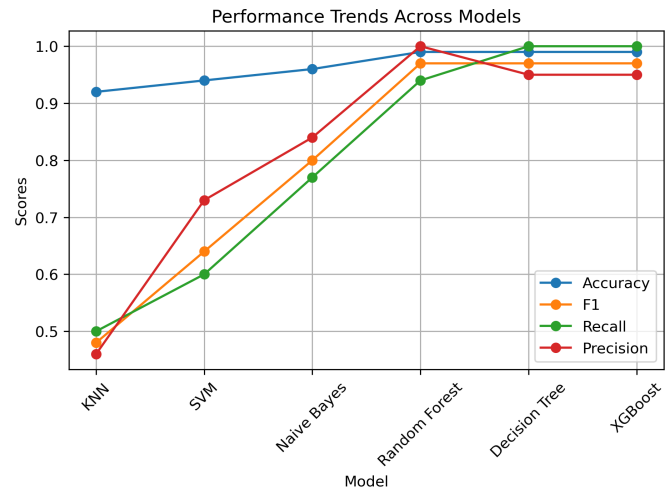


Fig. 9. A Line Graph of Classification Performance Trends

Fig. 9 shows performance trends, including Accuracy, F1 Score, Recall, and Precision, of various classification machine learning models. The trend analysis using the line plot shows that performance scores gradually improve from KNN to XGBoost. Notably, the Decision Tree has somewhat higher precision but poorer recall than Random Forest and XGBoost. This trade-off indicates that, while Decision Tree substantially reduces false positives, it may also overlook some good situations when compared to the other two models.

VI. CONCLUSION AND FUTURE WORK

This study proposes an integrated framework for stress detection using semi-supervised and supervised learning techniques, applied to physiological data collected during sleep. By addressing key challenges, such as limited labeled data and variability in individual stress responses, we successfully

conducted feature selection, mathematical transformations, and a self-training classifier to optimize the dataset. The results demonstrated the efficacy of using machine learning models in stress detection. Random Forest, Decision Tree, and XGBoost had shown outstanding performance and reliability in stress classification, reaching up to 99% accuracy and 97% F1-score. This emphasizes the promise of machine learning in creating dependable, non-invasive stress detection. It also helps to advance stress detection algorithms and emphasizes the need to address data quality and variability in physiological monitoring.

One important area for improvement is balancing the dataset to provide better representation of stress and non-stress states, which would reduce class imbalances and increase model robustness. Furthermore, investigate other datasets that incorporate data from more diverse population and wearable devices to study the models' generalizability across demographics and sensor types. Investigating data privacy techniques may potentially enable privacy-preserving collaboration across many datasets, increasing the adaptability and scalability of stress detection systems.

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